

Project Description

Build a 32-bit RV32I processor supporting these instructions first:

- **Arithmetic: ADD, SUB, ADDI**
- **Logic: AND, OR, XOR**
- **Shifts: SLL, SRL**
- **Memory: LW, SW**
- **Branches: BEQ, BNE**
- **Control flow: JAL, RET**
- **Comparisons: SLT**

Steps:

Learn SystemVerilog

Build ALU

Build Register File

Build a Single Cycle CPU

Pipeline the Single Cycle CPU

Implement Data forwarding, Hazard Detection, Branch Flushing etc.

Software Installed:

Icarus Verilog: To compile SystemVerilog Code and Run it.